Automatic DC-Offset Calibration for a Low-Cost Geophone Front End

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***Abstract*—This paper presents a foreground calibration method for a low-cost geophone analog front end in which stage-to-stage accumulation of amplifier and reference offsets can saturate intermediate nodes. The method sequentially measures and trims four ordered stages – a programmable-gain amplifier (PGA), band-pass filter (BP), summing stage (SUM), and low-pass filter (LP) – using existing on-chip conversion and filtering resources. Oscilloscope measurements on two independently assembled boards, with reported settings spanning the PGA's $\times 1$–$\times 50$ gain range, show ADC-facing LP sample-mean offsets of 0.4–3.9% of full scale across eight reported board/gain combinations, versus 2.9–45.8% under the fixed-reference baseline. The PGA is corrected only partially in some conditions; a forced-action diagnostic confirms that the normal deadband suppresses unnecessary DAC changes when stored operating points are already accepted, while downstream stages preserve the LP result. A four-stage run takes approximately 10 s typical (57.6 s worst case). Coherent multichannel surface-wave field records further demonstrate operation beyond the bench.**

***Index Terms*—Geophone, analog front end, DC-offset calibration, mixed-signal systems, PSoC**

I. Introduction

Multichannel surface-wave methods use distributed vibration measurements to estimate Rayleigh-wave dispersion and infer near-surface mechanical properties [1], [2]. Although their processing is well established, acquisition hardware remains a major source of cost and deployment complexity; compact wireless nodes based on inexpensive geophones and mixed-signal microcontrollers therefore enable denser, reconfigurable arrays [3]. Embedded interfaces already combine amplification, filtering, and programmable gain [4], but achieving low offset through higher-grade analog components raises the cost of every node. This work instead uses resources already present in the microcontroller to correct the operating point of a budget-constrained front end.

The system belongs to a low-cost, reconfigurable Internet-of-Things acquisition platform [5] based on SM-24 moving-coil geophones [6]. Its PSoC 5LP integrates programmable-gain and operational amplifiers, analog filters, voltage DACs, multiplexers, a delta-sigma ADC, a digital filter, DMA, and an Arm Cortex-M3 processor [7]. Time-sharing the ADC and digital filter between acquisition and calibration avoids a separate converter or processor, following digitally assisted analog design [8]. The tradeoff is that the internal analog blocks cannot be independently optimized for offset, drift, and flicker noise, as also reported for a single-chip PSoC potentiostat [9].

The geophone conditioning circuit is derived from the cascade correction and subtraction topology of Ma *et al.* [10]; mapped to the PSoC, accumulated DC errors prevented sufficiently aggressive gain settings and became particularly severe at the SUM output.

A continuously active DC servo, evaluated as an initial solution, proved impractical (Section V); the proposed method instead performs a finite foreground calibration before measurement and disconnects the controller afterward.

On-chip self-calibration is established. PSoC-Stat injects five IDAC levels and uses the ADC response to estimate TIA/ADC gain and offset [9]; Infineon's AN68403 measures a grounded PGA/ADC chain under an internal VDAC and stores numerical corrections in EEPROM [11]. Physical trimming has also been demonstrated for a single amplifier through body-bias calibration [12]. Here, instead of correcting only the ADC transfer characteristic or one amplifier, the loop adjusts the physical reference of every accessible analog node within the available DAC range and resolution.

The front end remains DC-coupled because the SM-24's usable band extends toward its mechanical corner: an input coupling capacitor would require a corner well below 1 Hz to avoid attenuating the targeted low-frequency content. AC blocking is therefore unsuitable, while substituting lower-offset external amplifiers would abandon the single-chip premise; per-stage trimming addresses offset without either compromise, unlike classical autozero/chopper techniques for offset and 1/f noise [13].

The contribution is threefold: 1) a per-stage analog calibration architecture using the existing PSoC multiplexer, ADC, digital filter, and four VDACs; 2) a foreground controller with deadband, anti-windup, slew limiting, stability detection, and one-LSB refinement that freezes its corrections before acquisition; and 3) validation on two independently assembled boards at settings spanning the PGA's $\times 1$–$\times 50$ gain range, showing bounded offsets without per-board or per-gain manual retuning.

II. Platform Design

The PSoC conditions and digitizes the geophone output in acquisition mode. In calibration mode, its measurement path, digital-filter coefficients, and DAC-control firmware form a temporary mixed-signal loop without changing the sensor input chain.

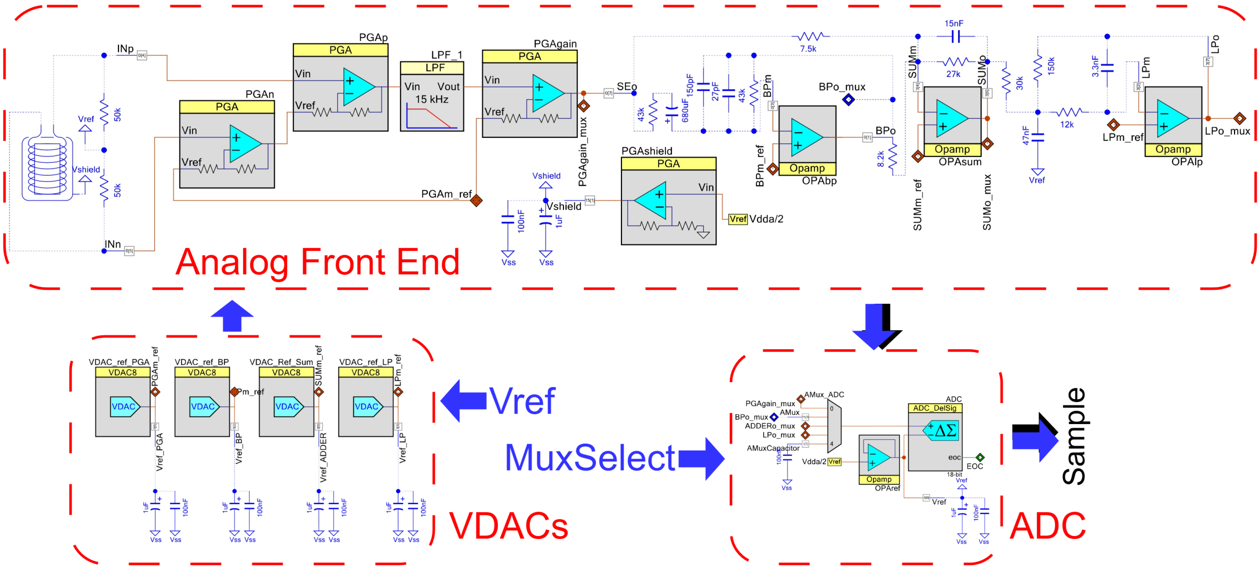

Fig. 1. Analog signal chain: four independent VDAC references trim the programmable-gain amplifier (PGA), band-pass filter (BP), summing stage (SUM), and low-pass filter (LP), with a shared ADC/multiplexer measurement path.

In the feed-forward signal chain of Fig. 1, an offset introduced at any stage can propagate through downstream gains toward the ADC input. The path is not a strict one-input cascade: BP follows PGA, whereas SUM combines a direct PGA branch with the BP branch before LP. The calibration therefore exposes and trims each stage separately rather than correcting only the final ADC value.

An analog multiplexer connects one stage output at a time to the shared 18-bit delta-sigma ADC, which serves as the error sensor; software mutual exclusion prevents simultaneous connections. A switched capacitor behind the selected multiplexer MOSFET combines with its on-resistance for limited RC anti-aliasing and isolates the amplifier from a direct capacitive load.

For the ith measured node, its slowly varying output can be represented as

$$y_i^{\mathrm{DC}} = \sum_{j<i} A_{ij} y_j^{\mathrm{DC}} + b_i + K_i u_i, \tag{1}$$

where A_{ij} is the DC coupling from an upstream node j (zero when no such path exists), b_i collects amplifier and reference errors, K_i is the signed transfer from the trim DAC to the measured node, and u_i is the DAC code. The multiple-input form covers both ordinary cascaded stages and nodes such as SUM.

BP ideally rejects DC, but SUM subtractively combines its output with a direct PGA branch. A mismatch between the DC operating points arriving through these two paths can therefore appear strongly at SUM even when BP itself has little residual offset. The first practical workaround was to adjust only the SUM reference; a later manual procedure adjusted every accessible node. The proposed method automates this per-node procedure. SUM-only correction can recenter that node but leaves earlier and later operating points uncontrolled, and it does not generalize to front ends without an explicit summing node.

The proposed architecture instead chooses each u_i in Eq. 1 independently so that every measured y_i^{DC} approaches its stage reference. In this differential front end, the physical target is $V_{\mathrm{ref}} \approx V_{\mathrm{DDA}}/2 \approx 2.5$ V; the ADC reports deviation from this reference, so the target corresponds to zero ADC counts rather than a physical 0 V node.

Eq. 1 is lower-triangular in node order: y_i^{DC} may depend on several earlier nodes but not on downstream ones. Processing PGA, BP, SUM, and LP in topological signal-flow order therefore places all upstream DC inputs at their targets before node i is trimmed, making a single forward pass sufficient (Section III).

III. Self-Calibrating Geophone Platform

A. Digital Hardware and Calibration Sequence

At boot, after a gain change, or on command, the firmware loads stored DAC codes and verifies the resulting offsets (Fig. 2). If all stages are within tolerance, it restores acquisition without a full run; otherwise it processes PGA, BP, SUM, and LP sequentially, resetting the FIR history and settling after each multiplexer switch. A watchdog and per-stage sample limit bound the sequence. During acquisition the controller is inactive, the multiplexer selects LP, and the VDAC codes remain constant, so no time-varying feedback enters the acquired signal.

The shared digital filter implements the geophone response during acquisition and switches to an aggressive low-pass estimator during calibration (Fig. 3); its acquisition coefficients and path are then restored. Successful DAC codes are stored in EEPROM as one record per PGA gain setting containing the gain code, valid-stage mask, four DAC codes, version, and CRC. A failed run never overwrites a valid record.

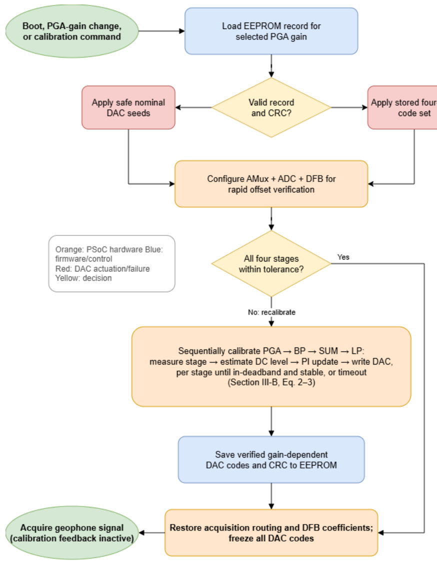

Fig. 2. Flowchart of the proposed foreground autocalibration algorithm, including stored-code verification, sequential per-stage correction, and transition to acquisition.

B. Control Law and Calibration Parameters

For each stage, the ADC reports the node voltage relative to $V_{\mathrm{ref}} \approx 2.5$ V. Thus, $\hat{y}_i[n] = 0$ denotes a physical node at V_{ref}, not at 0 V. The filtered differential measurement produces the ADC-domain error $e_i[n] = r_i - \hat{y}_i[n]$, with $r_i = 0$. The error is first mapped to the 8-bit DAC domain:

$$\varepsilon_i[n] = \mathrm{round}\left(\frac{e_i[n] V_{\mathrm{ADC}} N_{\mathrm{DAC}}}{N_{\mathrm{ADC}} V_{\mathrm{DAC}}}\right), \tag{2}$$

with $V_{\mathrm{ADC}} = 5000$ mV, $N_{\mathrm{ADC}} = 262144$, $V_{\mathrm{DAC}} = 4080$ mV, and $N_{\mathrm{DAC}} = 255$ in the implemented configuration.

The rounding in Eq. 2 makes $\varepsilon_i[n]$ an integer error in equivalent DAC codes. If the ideal operating point lies between two realizable codes, an acceptance band narrower than one output step can make the controller alternate indefinitely between them. The implemented deadband therefore scales with the DAC-to-node gain,

$$\Delta_i = \max(1, \lceil 1.2|K_i| \rceil), \quad (3) \qquad \bar{\varepsilon}_i[n] = \begin{cases} 0, & |\varepsilon_i[n]| \le \Delta_i, \\ \varepsilon_i[n], & \text{otherwise}, \end{cases} \quad (4)$$

where $\bar{\varepsilon}_i[n]$ is the error presented to the controller. The 1.2 factor provides a small margin over one realizable stage-output step, while the one-code floor preserves a finite acceptance region when $|K_i| < 1$.

The architecture requires only a measurable stage output and sufficient trim range; this prototype uses a positional PI law:

$$u_i^*[n] = u_{i,0} + s_i \left[\frac{k_{p,i}}{|K_i|} \bar{\varepsilon}_i[n] + \frac{k_{i,i}}{|K_i|} \sum_{q=0}^{n} \bar{\varepsilon}_i[q] \right], \tag{5}$$

Here $u_{i,0}$ is a feedforward code near the expected operating point, $s_i \in \{-1, +1\}$ is the injection polarity, and $|K_i|$ is the nominal stage movement per VDAC code in the scaled domain of Eq. 2. The fixed BP, SUM, and LP values are 2, 7.89, and 6, respectively; for PGA, $K_i = 1 - G$ and the injection polarity follows the active gain G. At $G = 1$, this trim path has no first-order authority, so the PGA code is held and calibration proceeds downstream; the normalization in Eq. 5 is applied only for $|K_i| > 0$. The feedforward codes were hand-set once on Board B at the power-up gain $\times 50$ and then used unchanged for Board A and all tested gains. With $k_{p,i} = 0$ for all stages, Eq. 5 reduces to the pure integrator parameterized by $k_{i,i}$ in Table I.

TABLE I
IMPLEMENTED PER-STAGE CALIBRATION PARAMETERS. DEADBAND Δ_i IS IN EQUIVALENT DAC CODES; SETTLING, LOCK, AND REFINEMENT ARE IN SAMPLES AT 2604 Hz.

Stage	K_i	Δ_i	$k_{i,i}$	Settle	Lock	Refine	Timeout (s)
PGA	$1-G$	gain-dependent	3×10^{-4}	512	512	1024	11.5
BP	2	3	5×10^{-4}	512	1024	1024	17.3
SUM	7.89	10	5×10^{-4}	512	512	1024	11.5
LP	6	8	5×10^{-4}	1024	1024	2048	17.3

The command is clamped to the VDAC range and slew-limited to one code per sample, with conditional integration for anti-windup. Convergence requires a fixed number of samples in the same quantized bucket; a final adjacent-code trial is retained only when it lowers the filtered error. At timeout, the best code found is held, the stage is flagged not converged, and the sequence advances.

Table I lists the implemented parameters. Settling uses 512 samples (0.20 s) for PGA/BP/SUM and 1024 (0.39 s) for LP; lock requires 512 or 1024 consecutive in-deadband samples, and refinement adds 1024–2048 samples. The listed timeouts are worst-case ceilings. The separate Board B $\times 4$ diagnostic used $\Delta_i = 1$ only to force controller action after normal verification had accepted all four nodes; it is not part of the $N = 10$ comparison in Section IV.

The 8-bit VDAC resolution meets this prototype's acceptance range. Applications requiring a smaller residual can trade added hardware for a finer step through a PSoC IDAC/resistor or a buffered topology [14], while retaining the same measure–estimate–correct sequence.

IV. EXPERIMENTAL VALIDATION

A. Prototype and Comparison Conditions

Two independently assembled SM-24 conditioning prototypes (Board A and Board B), each with four VDAC references, were tested under two conditions: *Default*, references hand-tuned once on Board B at gain $\times 50$ and reused unchanged; and *Proposed*, sequential four-stage calibration followed by frozen-reference acquisition. Default is therefore a fixed baseline calibrated for one board and operating point, not an uncalibrated circuit. The reported settings span the PGA's $\times 1$–$\times 50$ range and test whether per-board and per-gain retuning is necessary; Table II reports the resulting per-stage offsets.

B. Design Target: Residual DC Offset

The loop reduces physical DC offset rather than forcing exact mathematical zero. Its deadband, timeout, estimator, and gains balance residual error against convergence time within the available DAC, ADC, and FIR capabilities (Section III).

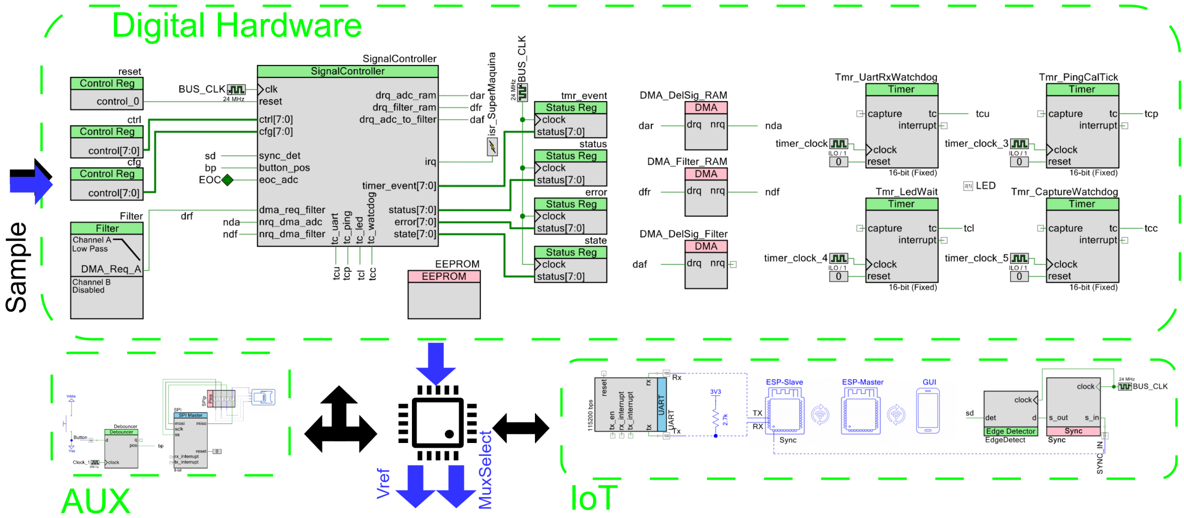

Fig. 3. Digital calibration hardware. The controller routes DMA transfers among the ADC, shared digital filter, and EEPROM and reconfigures the analog multiplexer. Black and blue arrows indicate acquisition and calibration paths, respectively.

TABLE II
RESIDUAL DC OFFSET AT EACH STAGE AS % OF THE ADC'S 0–5 V SPAN ($\pm$2.5 V ABOUT THE 2.5 V REFERENCE), REPORTED AS DEFAULT / PROPOSED MEAN $\pm$ SAMPLE SD (N=10 REPEATED RUNS PER REPORTED CELL; LOWER IS BETTER).

	Board A				Board B			
Gain	PGA	BP	SUM	LP	PGA	BP	SUM	LP
$\times 1$	$3.5 \pm 0.2/3.4 \pm 0.3$	$1.5 \pm 2.2/1.2 \pm 0.2$	$\mathbf{21.8 \pm 1.1}/0.8 \pm 0.2$	$\mathbf{45.8 \pm 3.1}/3.2 \pm 0.9$	–	–	–	–
$\times 4$	$2.4 \pm 0.1/2.1 \pm 0.3$	$4.9 \pm 0.3/1.0 \pm 0.2$	$6.7 \pm 1.0/1.0 \pm 0.3$	$\mathbf{42.6 \pm 4.2}/3.9 \pm 0.8$	–	–	–	–
$\times 16$	$2.4 \pm 0.1/1.9 \pm 0.2$	$3.8 \pm 0.9/1.5 \pm 1.5$	$4.9 \pm 1.5/1.4 \pm 1.5$	$\mathbf{29.6 \pm 6.8}/2.3 \pm 1.4$	$2.1 \pm 0.1/1.7 \pm 0.1$	$0.66 \pm 0.02/0.66 \pm 0.02$	$4.5 \pm 0.8/0.6 \pm 0.5$	$3.0 \pm 0.1/0.7 \pm 0.1$
$\times 32$	$1.0 \pm 0.2/0.5 \pm 0.2$	$7.0 \pm 0.4/1.0 \pm 0.5$	$7.4 \pm 0.3/0.6 \pm 0.2$	$\mathbf{12.4 \pm 1.3}/1.7 \pm 0.7$	$1.0 \pm 0.2/0.5 \pm 0.2$	$0.66 \pm 0.01/0.66 \pm 0.01$	$8.0 \pm 0.8/1.2 \pm 0.3$	$2.9 \pm 0.1/0.4 \pm 0.2$
$\times 50$	$1.2 \pm 0.3/1.0 \pm 0.5$	$1.0 \pm 0.7/0.9 \pm 0.5$	$0.4 \pm 0.2/\mathbf{0.8 \pm 0.6}$	$4.1 \pm 2.1/1.2 \pm 0.4$	$2.5 \pm 0.1/1.8 \pm 0.3$	$0.65 \pm 0.01/0.65 \pm 0.02$	$0.9 \pm 0.6/0.9 \pm 0.6$	$2.9 \pm 0.1/0.7 \pm 0.2$

Red: Default $\geq$10%FS. **Orange**: Proposed $\geq$ Default (Section V-A). Board B $\times 1/\times 4$: verification accepted all stages without a DAC change; the forced $\Delta_i = 1$ diagnostic at $\times 4$ (N=3) is excluded.

The ADC-facing signal is centered on a 2.5 V reference, so its 0–5 V input span corresponds to a $\pm$2.5 V excursion about that operating point. Every measured node must remain in its linear, unsaturated range; Section V-A examines partial PGA correction while the downstream LP offset remains small.

C. Measured DC-Offset Reduction Across the Gain Range

At each reported board/gain setting, a Tektronix TDS 2004B oscilloscope (4 channels, 60 MHz, 1 GS/s) measured the mean PGA, BP, SUM, and LP deviation from the 2.5 V reference over a 100 ms post-settling window under both conditions. Table II gives its absolute value as a percentage of the 5 V span, with N=10 repeated calibration runs per reported cell; the sample SD quantifies run-to-run repeatability under the same bench conditions, not unit-to-unit reproducibility. LP is the ADC-facing node; PGA, BP, and SUM expose upstream saturation or nonlinearity hidden by a final-output measurement alone. On Board B $\times 1$ and $\times 4$, stored-code verification checked all four stages and returned to acquisition without changing a DAC. At $\times 4$, forcing action with $\Delta_i = 1$ confirmed the acceptance logic, but the diagnostic was stopped at N=3 and is not a comparable Proposed condition (Section V-A).

Across the eight reported board$\times$gain cells, the LP sample mean spans 0.4–3.9%FS after calibration versus 2.9–45.8%FS under Default (Table II); individual Proposed LP observations span 0.1–5.2%FS. Board A's Default offset decreases with gain, whereas Board B remains closer to zero at its reported settings. Thus, within the tested boards and settings, calibration converts

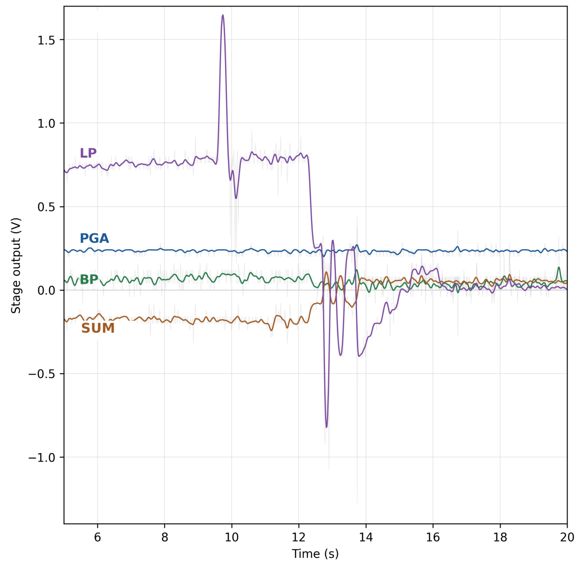

Fig. 4. Oscilloscope-derived PGA, BP, SUM, and LP outputs during a calibration run at PGA gain $\times 16$. Fixed references are active for $t \lesssim 9$ s; all stages converge by $t \approx 16$ s. The larger PGA residual is discussed in Section V-A.

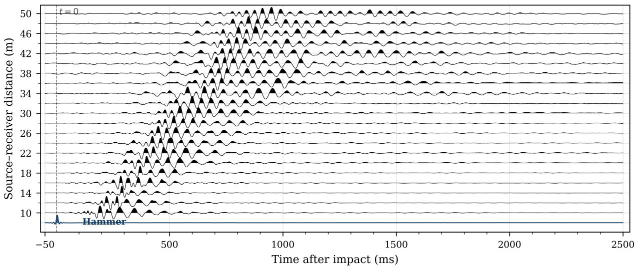

Fig. 5. Wiggle-trace field record acquired with the calibrated front end. The hammer trigger and first-arrival moveout are visible across the 10–50 m line.

a board- and gain-dependent operating point into a substantially smaller ADC-facing offset without manual retuning.

D. Field Use of the Calibrated Front End

Separate from the bench offset test, Fig. 5 shows wiggle traces from a field surface-wave record over a 10–50 m line. First-arrival moveout is resolved at every offset using higher PGA settings at far offsets; this demonstrates field operation but is not a controlled Proposed-vs-fixed-reference comparison.

V. Discussion

The method trades component precision for system-level measurement and digital assistance, unlike precision front ends characterized near the sensor's Johnson-noise limit [15]. Its benefit is not that the ADC output has zero mean – a software subtraction could do that – but that the DACs modify the physical operating points before large downstream gains apply, so the correction is agnostic to the error source, provided the stage is measurable and the DAC has sufficient authority and range.

The foreground method replaced a tested continuous DC servo: a low corner settled impractically slowly, a higher corner interfered with the signal, and each gain change restarted the transient, matching the corner-frequency/settling tradeoff reported for DC-servo compensation [16]. The proposed controller disconnects during acquisition, holding the DAC outputs constant and adding no time-varying feedback or servo poles. EEPROM avoids most full runs while the errors remain repeatable; if aging, temperature, or another condition moves a stored point outside tolerance, verification rejects it and recalibrates at boot, after a gain change, or on command (Section III). Remaining limitations are finite DAC resolution, no correction of noise or distortion, and unvalidated convergence under sustained vibration or thermal drift.

A. The PGA Stage: A Partial Result, Explained

Unlike BP, SUM, and LP, the PGA stage's post-calibration offset is not consistently much smaller than Default: the mean $|\text{offset}|$ falls only modestly at low gain and by roughly half at best. The observed behavior is consistent with two firmware choices. First, the deadband Δ_i (Section III) is an acceptance margin, not a physical limit. On Board B at $\times 4$, stored-code verification found all four stage outputs acceptable and returned without moving a DAC. Repeating the diagnostic with $\Delta_i = 1$ caused the controller to act, confirming that the normal no-action result followed the implemented acceptance logic; the diagnostic was stopped at $N{=}3$ and was not included in Table II. Second, the 128-sample FIR estimator trades residual resolution against convergence time; whether a longer estimator materially improves the remaining PGA offset was not tested. The diagnostic demonstrates available trim action but does not establish the ultimate PGA residual floor; neither limitation changes the measured LP offset reduction.

The orange cell in Table II occurs at Board A SUM $\times 50$, where the fixed-reference mean offset was already 0.4%FS and the Proposed mean remains 0.8%FS. The controller is designed to enter an acceptance band, not to guarantee a lower sample mean in every repeated cell, so this small reversal does not contradict the downstream offset result.

Fig. 4 shows the sequential PGA $\rightarrow$ BP $\rightarrow$ SUM $\rightarrow$ LP correction: each measured output steps when addressed and then holds. The four-stage sequence takes approximately 10 s typical; the Table I timeouts sum to 57.6 s only if every stage times out, which was not observed.

The two boards have markedly different Default operating points, yet both exhibit small LP mean offsets at their reported settings. Feedforward values were tuned only on Board B at $\times 50$; Board A received no manual adjustment and still converged across its tested gains. This supports reuse of one nominal initialization in a dense network, with the loop absorbing unit- and gain-dependent variation. The $N{=}10$ runs quantify within-cell repeatability, but two boards cannot establish population-level scaling; more units and cross-temperature trials are required.

VI. Conclusion

A per-stage foreground calibration method was implemented in a low-cost PSoC geophone front end using four VDACs, the existing analog multiplexer, acquisition ADC, and digital filter. It trims the physical operating point of each stage, freezes the resulting codes during acquisition, and verifies stored gain-dependent corrections before reuse.

Across eight reported board/gain cells on two boards, collectively spanning $\times 1$–$\times 50$, the ADC-facing LP sample-mean offset was 0.4–3.9%FS after calibration versus 2.9–45.8%FS under the fixed-reference baseline, without per-board or per-gain manual retuning. A forced-action diagnostic confirmed that stored settings inside the normal acceptance band are intentionally left unchanged. Calibration takes approximately 10 s typical (57.6 s worst case). Field records demonstrate operation over a 10–50 m line; population scaling, temperature dependence, VDAC-reference noise, and a controlled field comparison remain to be measured.

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